

An Evaluation of Persons with Disabilities Employment Status in the United States

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Abstract

On July 26, 1990, 41st President of the United States, George H.W. Bush signed the Americans with Disability Act into law. Civil rights activists concluded those who were born with a disability faced serious discrimination as clients who freely used public space and businesses. The activists also concluded the disabled were discriminated against while seeking employment. This paper discusses several assessments of the current state of employment as well as a categorical analysis of different disability types with regards to employment status.

Keywords: Persons with Disabilities, Vocational Rehabilitation Services, Labor Force, Logistic Regression, Principal Component Analysis, Simple Exponential Smoothing, Linear Discriminant Analysis, Multicollinearity

Objective

The objectives are to explore data that oversees employment status of the civilian population in the United States. There are a few U.S. government sources that have resources to generate statistical models. Forecasting methods can also be utilized to generate a forecast for number of persons in the workforce for 2023 through 2030. A conclusion can be drawn on the magnitude of employing Persons with Disabilities and employment of non-disabled persons.

U.S. Bureau of Labor Statistics¹ provides data that can be used for the objectives. This federal government agency provides published electronic data that is released to the public domain. These data sets contain raw quantitative data that describe various options of the labor force with total numbers reported on a monthly basis, from the years 2009 to 2022. The reported number of persons is numeric and counted as persons who are sixteen years or older and has a disability or no disability. Several variations of the data sets can be selected and obtained. These variations include Total, 16 years and over, Men, 16 to 64 years, Women, 16 to 64 years, and Both sexes, 65 years and over. The data sets that can be obtained are reportedly categorized as Persons with a disability and Persons with no disability. Additional variations exist; employment-population ratio and number employed as well as a few others can be found.

Another objective is discussed. This objective observes the classification of a disability and the employment status. This involves a data set from one of the many Vocational Rehabilitation networks throughout the United States. A conclusion of effectiveness of VR program can be drawn. Iowa Vocational Rehabilitation Services² has an interesting data set. This data set is a non-federal data set that was released to the public domain. This data set provides case information from IVRS and contains the information of cases that were opened from the year 1990 onwards. Cases are reported to have closed up to the year of 2022. The data set reports that 54,593 cases were opened and closed from 1990 to 2022, all in the state of Iowa.

A person with a disability is described as a person who has a medical history of being blind, visually impaired, deaf, or hearing impaired. The person who has a disability also can have mental problems that are diagnosed as serious conditions. Others who may have a disability can

¹ United States Bureau of Labor Statistics, Employment status of the Civilian population by sex, age, and disability status, not seasonally adjusted. 2009-2022, Raw data (Washington DC: U.S. Bureau of Labor Statistics, May 17, 2023).

² Iowa Vocational Rehabilitation Services. Closed Iowa Vocational Rehabilitational Cases. 1990-2022. Raw data (Iowa: Iowa Vocational Rehabilitation Services. June 3, 2023).

have serious difficulty walking, bathing or doing errands alone. Disabilities are described as a serious limitation in the physical, mental, or emotional well-being of a person and cannot be corrected with a doctor's prescription. The IVRS data set classifies five major disabilities as deaf and hard of hearing, visual and communication, physical, psychosocial mental and cognitive. Each of the 54,593 cases opened and closed are provided with one of these five classifications of a disability. Upon the closure of each case, the employment status is reported.

Literature Review

Hiring and working with people who have disabilities is an interesting topic viewed throughout the world. In 1986, a United States government agency utilized a purposeful study to identify that those who have a disability were more likely to be unemployed. As a result of the study, showing the disabled were discriminated against when seeking employment and other forms of public life entities, the Americans with Disabilities Act was signed into law in 1990. Persons with Disabilities were not protected from discrimination under the initial Civil Rights Act of 1964; the legislation that was signed into law in 1990 makes discrimination against PwD illegal in the workplace and other areas. With the current corona virus pandemic, an unfortunate overview of PwD into the workplace has resulted. Giving qualified people who have disabilities a job, without discrimination, is an ethical business decision that can provide a positive workplace. For a definite analytics overview, models will be drawn from specific entities where data was obtained.

A global social issue exists. Lack of implementing the ADA or other friendly disability legislation is widespread, throughout much of the developed world. Approximately 15 to 20 percent of the world population³ is disabled; over one billion people. Even the twenty-first century is not so friendly to persons with disabilities. Data has shown about 500 million PwD are at the age to join the labor force, world-wide. However, three decades after the signing of the ADA and other world-wide disability friendly legislation, discrimination⁴ is still a barrier for PwD to gain employment. Data collected demonstrates that the employment rate for PwD globally⁵ is only 48.7 percent; much lower than the 72.5 percent employment rate enjoyed by people who do not have a disability. Several global organizations seek to obtain a fair playing workplace for all by 2030; demanding equal pay of equal value, decent work and employment opportunities for all men and women, including PwD.

The United States also has a wide statistical contrast in the labor force. Of the more than 18 million⁶ working-age disabled people in the United States, the percentage of employed PwD is only 33 percent. In contrast, the percentage of non-disabled people enjoys a 76 percent employment rate. This demonstrates the employment gap between PwD and non-disabled people in the United States. Disabled employees are paid one-third less than their non-disabled

³ M Ángeles Jurado-Caraballo, Cristina Quintana-García, and Mercedes Rodríguez-Fernández, "Trends and opportunities in research on disability and work: An interdisciplinary perspective," *Business Research Quarterly*, 25, no. 4 (2022): 296 <https://doi.org/10.1177/2340944420972715>.

⁴ M Ángeles Jurado-Caraballo and Cristina Quintana-García, "Trends and opportunities in research," 296.

⁵ M Ángeles Jurado-Caraballo and Cristina Quintana-García, "Trends and opportunities in research," 296.

⁶ Sally Lindsay et al, "A systematic review of the benefits of hiring people with disabilities," *Journal of Occupational Rehabilitation* 28, (2018): 636, <https://link.springer.com/article/10.1007/s10926-018-9756-z>.

coworkers and this demonstrates the unethical approach to hiring PwD⁷. The initial phase of the corona virus pandemic has set a negative impact on PwD in the workplace, as well. Some have dealt with lost wages and job loss due to employers having constant fear of exposure to COVID-19. General labor force statistics show that 20 percent of disabled persons who were employed lost their jobs from March 2020 to April 2020 and only 14 percent of non-disabled persons lost their jobs during that same time period⁸. Two-thirds⁹ of PwD want to work but cannot find work and they need to be included and accepted as a serious value of the labor force.

Despite the negative social issue of PwD in the workplace, companies who do hire persons with disabilities have gained a vast amount of benefits. Research articles, from 1997 to May 2017 have systematically demonstrated success within the companies that chose to hire PwD¹⁰. The benefits these companies gained included improvements in profitability and brought a competitive advantage. Most importantly, the positives brought a diverse work culture as well as ability awareness. Of the bill that was signed into law, by 41st President of the United States George H.W. Bush, several entities were against enacting the law. Initially, those who were opposed to the bill had persistent belief that the disabled would cost more to employ due to the reasonable accommodations that the company would need to purchase. A study that was published at the Department of Occupational Sciences and Occupational Therapy at University of Toronto¹¹ shows companies purchasing the reasonable accommodations are going to retain their workers and gain benefits that far outweighs the expenses of the accommodations.

Methods

Descriptive statistics were developed. A quantitative data set was computed for the mean, median, range, standard error, standard deviation and mean absolute deviation. The descriptive statistics also included finding the min and the max as well as computing skewness. Table 1

TABLE 1. *Descriptive Statistics(Quarterly employment-population ratio data set)*

	vars	n	mean	sd	median	mad	min	max	range	skew	se
Year	1	28	2015.5	4.11	2015.5	5.19	2009	2022	13	0	0.78
Q1	2	28	41.31	23.45	41.63	34.4	16.3	65.87	49.57	0	4.43
Q2	3	28	41.41	23.46	39.47	33.6	16.77	66.33	49.56	.01	4.43
Q3	4	28	41.58	23.54	41.08	34.35	17.27	66.4	49.13	0	4.45
Q4	5	28	41.7	23.45	42.42	34.25	17.03	66.63	49.6	0	4.43
Dummy	6	28	.5	.51	.5	.74	0	1	1	0	.1

shows the descriptive statistics for the quarterly employment-population ratio data set. The data set of

the descriptive statistics, found on Table 1, were developed by taking the employment-population ratio¹² data sets for both disabled and non-disabled categories and computing the

⁷ National Governors Association, *Governors' Role in Promoting Disability Employment in COVID-19 Recovery Strategies*, March 23, 2021, https://www.nga.org/wp-content/uploads/2021/03/SEED_Memo.pdf.

⁸ National Governors Association, *Governors' Role*, 1-2.

⁹ Sally Lindsay et al, "A systematic review of the benefits," 636.

¹⁰ Sally Lindsay et al, "A systematic review of the benefits," 634-655.

¹¹ Sally Lindsay et al, "A systematic review of the benefits," 639-646.

¹² U.S. Bureau of Labor Statistics, *Employment-Population Ratio age 16 and over, 2009-2022*, Raw data.

quarterly employment-population ratio. The quarterly employment-population ratio was computed by finding the average employment-population ratio for every three months and then rounding to two decimal places. This produced a data set that generated quarterly data from 2009 to 2022. A dummy variable was implemented. This dummy variable is a binary variable and a 1 is the disabled quarterly employment ratio and a 0 represents the non-disabled quarterly employment ratio. After the quarterly employment-ratio data set was developed, descriptive statistics were computed.

A second set of descriptive statistics were computed. These descriptive statistics, shown in Table 2, are dedicated to a data set where a yearly average number of employed persons in the labor force were taken. The data set from the U.S. Bureau of Labor Statistics¹³ contained the

TABLE 2. Descriptive Statistics(Yearly average of Employment Population data set)											
	vars	n	mean	sd	median	mad	min	max	range	skew	se
Year	1	14	2015.5	4.18	2015.5	5.19	2009	2022	13	0	1.12
Non-disability Average	2	14	142923.07	6207.93	143041.0	7587.21	134125	151680	17555	-0.05	1659.14
Disability Average	3	14	5440.36	564.70	5273.5	462.57	4861	6956	2095	1.25	150.92

total employed numbers in the labor force. The average of the twelve months from the years 2009-2022 was taken to develop the yearly average of employment population data set.

Logistic Regression is a parametric method known for computing probabilities. This parametric method involves a dependent variable that belongs to a class or category as well as the independent variables being categorical. Logistic regression usually involves qualitative data. Logistic regression produces probabilities where the dependent variable is concerned with a specific class or category. The dependent variable has a binary coding of a 1 or a 0 and is modeled using the logistic function. Table 3 displays the variables included in the logistic regression model. Table 3 involves data from the IVRS¹⁴ data set. Binary coding was implemented via a dummy variable. The logistic regression model will generate probabilities dedicated to the dependent variable of Employed Closure with multiple tests. This will uncover the known disability type and the probability of cases that were unsuccessful at finding employment based on the disability type. The independent variable, Gender, can also declare some association between the dependent variable. Gender could be removed from the model if

TABLE 3. Variables in Logistic Regression model	
Variable	Description
Employed Closure	Upon the case being closed, if employed a “Yes” is reported. If unemployed after cased closed, a “No” is reported. Implementing binary coding as a 1 for Yes and a 0 for No
Disability Type	The five disabilities reported on the IVRS data set: physical, psychosocial mental, cognitive, visual & communication and deaf & hard of hearing
Gender	Did Not Self-Identify, Male or Female reported via each case

not proven significant with the dependent variable.

Simple Exponential Smoothing is one of several different forecasting methods. This method is also known as First Order Exponential Smoothing. Simple Exponential Smoothing follows a special form of weighted parameters.

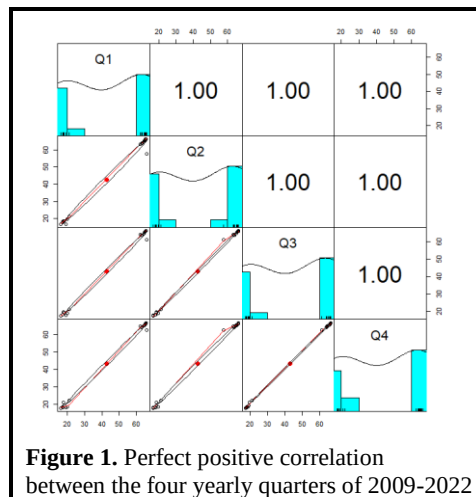
¹³ U.S. Bureau of Labor Statistics, Employed age 16 and over, 2009-2022, Raw data.

¹⁴ Iowa Vocational Rehabilitation Services. Closed Iowa Vocational Rehabilitational Cases. 1990-2022. Raw data.

The weights that make up this model are an exponential average. This model takes into perspective weights that account for past observations. The past observations are penalized with respect to the geometrically decrementing occurrences throughout the past data. Simple Exponential Smoothing can generate effective forecasts, if the time series analysis does not have a trend or does not concern seasonal data. SES is best used for forecasting a time series that rather fluctuates to a certain extent. This form of forecasting generally follows a vital formula. The formula is $A_t = \alpha x_t + (1 - \alpha)A_{t-1}$ and A_t is described as the smoothed mean of a time series. The constant α is a smoothing parameter that is implemented as $0 < \alpha < 1$.

U.S. Labor Force Statistics¹⁵ data was prepared. The data set showed 12-month totals of workers in the labor force from 2009 to 2022. A yearly average was taken and a data set was constructed. Table 2 shows these variables as well as the descriptive statistics for the data set. With the yearly average computed and the time series not concerning seasonal data or a trend, simple exponential smoothing seemed to be an effective option to forecast yearly average workers in the labor force for 2023 to 2030. Two categories will be constructed; the disabled workers in the labor force will be forecasted as well as the non-disabled workers in the labor force.

Principal Component Analysis is an unsupervised statistical learning method. PCA is one of many methods that can be used to deal with highly correlated variables. Principal components are used to find the variability, with the use of components, by developing a small number of representative variables that describe a large measurement of the variability of the original set of observations. Principal components are then demonstrated in a feature space. When the feature space is developed, directions are drawn as vectors. The directions of the vectors in the feature space distinguishes how closely related the original data is; defining variability. When developing a PCA, the independent variables only contribute to the model.



An establishment of PCA is to be shown with the quarterly employment-population ratio data set. This data set was initially discussed in Table 1. Table 1 showed the descriptive statistics of this data set. This data set was computed by finding the average employment-population ratio of every three months of the year. The years included 2009 to 2022. To investigate on whether PCA should be developed, the measurement of correlation was taken between each of the four

¹⁵ U.S. Bureau of Labor Statistics, Employed age 16 and over, 2009-2022, Raw data.

independent variables. The four independent variables are Q1, Q2, Q3 and Q4. Figure 1 shows the correlation between the four independent variables. The correlation is a perfect positive correlation and this shows that PCA is an approach that can be taken to distinguish these variables.

Linear Discriminant Analysis is another classification method. This supervised method uses Bayes' Theorem. Linear Discriminant Analysis typically works best where there are more than two classes that define the dependent variable. However, this method is not limited to using more than two classes in dependent variable. LDA approximates Bayes classifier. If the independent variables are greater than one, a multivariate Gaussian distribution is assumed.

PCA can be applied as a preprocessor method. Using PCA as preprocessor can invalidate the issue of multicollinearity. Using PCA and then generating an LDA model, with the dependent variable being the dummy variable and independent variables being the two principal components that hold most of the variability, can make predictions on which category of the employment-population ratio has the greatest chance of incrementing in the near future. The dummy variable will be re-inserted into the data set after PCA. The dummy variable describes 1 as being the disabled employment-population ratio and 0 as non-disabled employment-population ratio.

Results

The logistic regression model was initially preformed with Employed Closure assigned as the dependent variable and Gender as an independent variable with disability type. The p-value for Gender being Male is 0.116 and the p-value for Gender being Female is 0.138. The large p-value for Gender indicated that there is no real association between Gender and the dependent variable of Employed Closure. Gender was removed from the model to introduce a better fit.

Logistic regression model was re-established. The model was created with the dependent variable being Employed Closure and the independent variable being Disability Type. In this re-established model, three of the five disability types have the highest level of significance. These three disability types are Deaf & Hard of Hearing, Physical and Psychosocial Mental. Predictions were then developed. The established predicted probability of an increase in employment status after the VR case closed was set to greater than a 0.5 probability. A confusion matrix, in Table 4, was produced. The confusion matrix provides information that describes the number of observations that were correctly and incorrectly classified. This is a measure of the accuracy of the logistic regression classifier. The model accurately predicted that upon closure of IVRS cases, 21,121 cases ended up with the client having a job when case was closed. The model also accurately predicted 10,293 cases ended up with the client not having a job at the closure of the client's IVRS case. The model has made 31,414 correct predictions and 23,179 incorrect predictions. Logistic regression has predicted the accuracy of IVRS closed cases 57.5 percent of the time. The training error rate has been established as 42.5 percent. Based upon the data inside this model, 32.8 percent of clients who move through the IVRS will not be employed upon the closure of the client's case. The model further predicts 67.2 percent will be employed after the client participates and goes through IVRS.

TABLE 4. *Confusion Matrix*

	NO	YES
NO	10293	8264
YES	14915	21121

The logistic regression model was expanded. A training set was assigned. A sample was taken from the original data set to create the training set. Five tests were created. The training set consisted of 43,784 observations. The tests filtered each disability type. Probabilities were

TABLE 5. Logistic Regression Predictions of Disability type (Employed after case closed)

Physical	51.9%
Psychosocial Mental	55.1%
Cognitive	59.8%
Visual & Communication	61.2%
Deaf & Hard of Hearing	75.2%

created for these disability types. With a 0.5 probability threshold, table 5 shows the disability type where IVRS successfully was able to report employment status as being employed upon the closure of the client's case.

Simple Exponential Smoothing is developed to forecast employed labor force numbers. The establishment of the time series of Disabled Workers, Figure 2, in the U.S. from 2009 to 2022 shows a behavior of no real trend but seems to be incrementing though the last year of 2022. This could indicate a growing trend. However, most of the past observations of this time series seem to be fluctuating. Figure 2 displays the time series with simple exponential smoothing forecasts for disabled workers

Simple Exponential Smoothing is

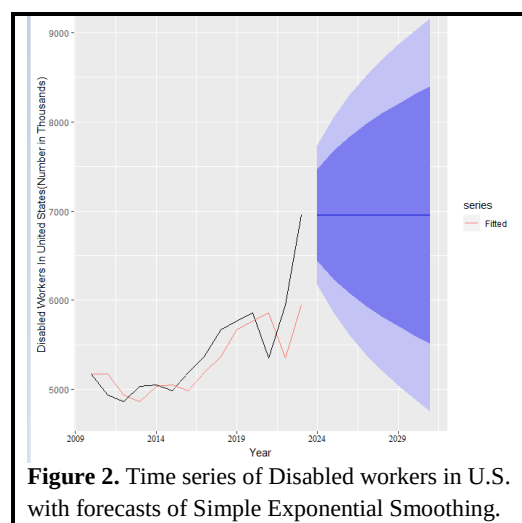


Figure 2. Time series of Disabled workers in U.S. with forecasts of Simple Exponential Smoothing.

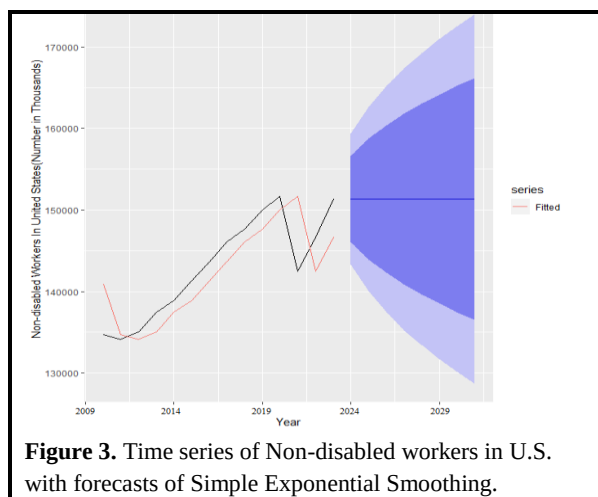


Figure 3. Time series of Non-disabled workers in U.S. with forecasts of Simple Exponential Smoothing.

employed in the labor force. The black line inside Figure 2 demonstrates the historical time series. The red line

demonstrates Simple Exponential Smoothing. The corona virus pandemic is a historical event that dropped the labor force numbers significantly, in 2020. Figure 2 shows that there are more disabled workers employed in the labor force than the last decade. There are more disabled workers employed in the labor force, today, than even before the corona virus pandemic, according to the time series in

Figure 2. The forecasts of the simple exponential smoothing method are displayed in Table 6. Table 6 shows that there is potential to be a number between eight to nine million disabled persons employed in the labor force by 2030, if the increase enjoyed in 2022 is to continue. On the lower scale of Table 6, Lo 80 and Lo 95, the number of disabled workers employed could make a return to 2011 numbers and be between four and five million by 2030. The number of disabled persons employed will need to take a downward spiral to reach the forecasts computed in Lo 80 and Lo 95.

TABLE 6. Disabled Workers Simple Exponential Smoothing Forecast (In Thousands)

Year	Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2023	6955.899	6445.937	7465.861	6175.980	7735.819
2024	6955.899	6234.740	7677.059	5852.981	8058.817
2025	6955.899	6072.678	7839.121	5605.129	8306.670
2026	6955.899	5936.052	7975.747	5396.177	8515.622
2027	6955.899	5815.681	8096.118	5212.085	8699.713
2028	6955.899	5706.857	8204.942	5045.653	8866.145
2029	6955.899	5606.782	8305.017	4892.603	9019.196
2030	6955.899	5513.635	8398.164	4750.146	9161.652

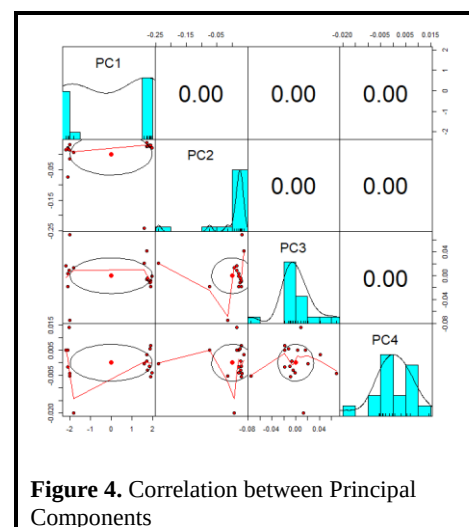
The non-disabled workers employed in the labor force were also forecasted. The black line in Figure 3 displays the historical time series of non-disabled workers employed in the labor force. The red line displays the simple exponential smoothing results. Figure 3 shows a five year incrementing trend, between 2014 and 2019. However the first five years of this time series appear to be fluctuating around a base. The corona virus pandemic has created a downward trend from 2019 to 2021. In 2020-2021, the number employed lost, during the initial stages of the corona virus pandemic seems to have fully recovered to pre-pandemic numbers. Simple exponential smoothing was established, in a red line, in Figure 3. By closely examining Figure 3, the forecast predicts that between 165 and 175 million non-disabled workers will be employed by 2030. However, the number employed might see another downward spiral. For forecasting a decrementing time series, the numbers projected appear to be between 130 and 135 million non-disabled persons employed in the workforce by 2030.

The training errors for the forecasting models, displayed in Figure 2 and Figure 3, are displayed in Table 7. With the Mean Absolute Percentage Error being less than 10 for both models, Simple Exponential Smoothing appears to be an acceptable forecasting method for these data sets. The lower the MAPE, the more accurate the forecasts appear. The non-disabled data, Figure 3, appears to have a lower MAPE and more accurate predictions are expected than the Disabled data of Figure 2. The Root Mean Square Error is a measure of fit. A lower RMSE means a better fit. Disabled employed workers forecast has a better fit than the non-disabled forecasting fit.

TABLE 7. Simple Exponential Smoothing Training Errors							
	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Disabled	127.29	368.41	253.29	1.93	4.37	0.93	0.15
Non-Disabled	746.65	3783	3031.94	0.48	2.12	1.09	-0.05

Principal Component Analysis was implemented. The quarterly employment-population ratio data set was applied with PCA. To observe on whether an issue with multicollinearity could be developed, on the further analysis of Linear Discriminant Analysis, the correlation was measured between the four principal components, shown in Figure 4. With a correlation of zero developed, multicollinearity should not become an issue.

TABLE 8. The Principal Component Loading Vectors(Quarterly Employment-Population ratio data set)		
	PC1	PC2
Q1	0.4996468	-0.76265641
Q2	0.4998362	0.62922556
Q3	0.5003036	0.14892078
Q4	0.5002132	-0.01590614



Principal component loading vectors are established in Table 8. The weights of the loading vectors of the first principal component appear to be distributed at approximately the same weight throughout the four variables. Most of the weight of the second principal component is placed on Q2. With more weight on Q2 in the second component, the second quarter of the employment-population ratio data set has the two greatest ranked loading vectors located at (0.4998362, 0.62922556). This indicates that Q2 has the greatest emphasis on employing workers in The United States. The months of April, May and June, the second quarter of the year, tends to have a greater emphasis on hiring workers than all of the other quarters, combined. Figure 5 displays the first two principal components with their loading vectors. Figure 5 is known as a biplot. According to figure 5, Q3 and Q4 demonstrate a close correlation. The variables Q2 and Q1 appear to be less correlated with each other as well as being less correlated with Q3 and Q4. This indicates that Q3 and Q4 have the same emphasis on hiring; Q2 has the greatest emphasis and Q1 has the least emphasis.

Table 9 shows the significance of the components. The first principal component explains 99.86% of the variance. The second principal component explains .11% of the variance. With 99.86% of the variance found in PC1, the other 3 components only combine to share .14 percent of the variance.

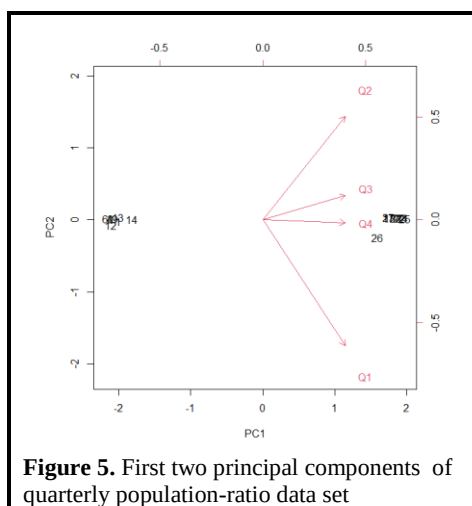


Figure 5. First two principal components of quarterly population-ratio data set

TABLE 9. Importance of components				
	PC1	PC2	PC3	PC4
Standard deviation	1.9986	0.06734	0.02991	0.007344
Proportion of Variance	0.9986	0.00113	0.00022	0.000010
Cumulative Proportion	0.9986	0.99976	0.99999	1.000000

Linear Discriminant Analysis was utilized on the training data after PCA was established on the training set. A dummy variable was formatted into the training set. The dummy variable describes a 0 as non-disabled and 1 as disabled. The model was fit as the dependent being the dummy variable and the independent being

PC1 and PC2. The prior probability is .4545 and .5454. The training set established that 45.45% of all observations are years to which category of non-disabled was established over the opposing category. This corresponds to years during which employment-population ratio maintained a presence. For the category of disability, the group means, in Table 10, is negative and group mean is positive for non-disabled category. This shows the gap

TABLE 10. Group Means		
Variable	PC1	PC2
0	1.810244	0.024108547
1	-2.080425	-0.005228626

between these categories with respect to employment-population ratio.

Predictions were created. A 50% threshold was applied to the posterior probability. An increase in the employment-population ratio for the category of being disabled is predicted to increase for the next 9 years. Considering that the employment-population ratio for the category of disabled has been incrementing since 2009 and employment-population ratio for the category of non-disabled has been rather constant since 2009, this model appears valid.

Limitations

A limitation exists. The data obtained for several of the assessments could only be gathered from the years 2009 to 2022. The ADA was signed into law in 1990; having thirty or more years of data could offer more significance than the less than fifteen years of data available. Discoveries could be further established on how the employed population responded to the severe recessions of the 2000s as well as other historical economic events.

Another limitation exists. The use of an unsupervised method, principal component analysis, invokes a much more challenging setting. No clear goal can exist with an unsupervised method and these methods are often established upon an exploratory data analysis. Validating and assessing results from an unsupervised method has no clear intention; no clear answer exists as the setting is unsupervised.

Conclusions

More than thirty years have passed after the official signing of the legislation that protects disabled persons from discrimination in the labor force as well as other public entities. Several inferential models were discussed that created assessments based upon employment data. Today, there are more disabled workers employed in the labor force than the last decade. This can contribute to the fact that the corona virus pandemic has opened the doors to a more flexible work place; making jobs more bearable for those who have a disability. This could seemingly conclude on the behavior of the increase of the last two years of disabled workers employed into the labor force; validating that more disabled workers are employed, today than before the historical event of the pandemic.

With eighteen million disabled persons, at the age to join the labor force, in the United States, there is room for improvement. Of the eighteen million disabled people who are at the age to join the labor force, more than eleven million need to be employed to reach the same magnitude as non-disabled persons. The non-disabled employment-population ratio is one-third of that for disabled workers. With simple exponential smoothing forecasts computed, the number of eleven million disabled persons employed in the labor force will not be expected to be reached by 2030. However, the disabled persons employed have enjoyed an increase in the last few years.

Vocational rehabilitation networks across the country can continue and try to make improvements. Deaf & Hard of Hearing persons have a 75 percent chance of being employed after moving through Iowa Vocational Rehabilitation Services, after the case is closed, but for persons with a physical disability that figure drops to approximately 52 percent. Improvements could be made but the situation between IVRS and the disabled person could rather depend on other variables than disability. These other variables could be educational attainment, attitude towards employment and the values of the individual.

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Appendix

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#####PCA + LDA#####
#####PCA####
#working directory
setwd(choose.dir())
library(MASS)
library(psych)
library(ISLR)
EmpQ <- read.csv("Q.csv",header=TRUE, sep=",")
dim(EmpQ)
describe(EmpQ)
EmpQ
set.seed(555)
Sampled<-sample(2,nrow(EmpQ), replace=TRUE, prob=c(0.5,0.8))
trainingSet<-EmpQ[Sampled==1,]
testingSet<-EmpQ[Sampled==2,]
trainingSet
testingSet
names(testingSet)
#Dummy Variable
#1 Disabled
#0 Non-disabled

pairs.panels(testingSet[,-1][,-5],gap = 0,
bg = c("red", "black", "blue")[trainingSet$Dummy],
pch=21)
```

```
testingSet
## examine the variance
apply(testingSet[,-1][,-5],2,var)
# examine the mean
apply(testingSet[,-1][,-5],2,mean)
pr.out=prcomp(testingSet[,-1][,-5],
center=TRUE,scale=TRUE)
attributes(pr.out)
names(pr.out)

pr.out$center

pr.out$scale
#standard dev, loadings
print(pr.out)
#
summary(pr.out)
pr.out$rotation
dim(pr.out$x)

names(trainingSet)
pairs.panels(pr.out$x,
             gap=0,
             bg = c("red",
                    "yellow", "blue")
[trainingSet$Dummy],
             pch=21)
```

```
#####
```

```
biplot(pr.out,scale=0)
```

```
pr.out$rotation=-pr.out$rotation
```

```
pr.out$x=-pr.out$x
```

```
biplot(pr.out,scale=0)
```

```
##access standard deviations
```

```
pr.out$sdev
```

```
#variance explained by each principal component
```

```
pr.var=pr.out$sdev^2
```

```
pr.var
```

```
#proportion of variance expained by each principal component
```

```
pve=pr.var/sum(pr.var)
```

```
pve
```

```
round(pve,digits=5)
```

```
plot(pve, xlab="Principal Component",ylab="Proportion of variance Explained",  
ylim=c(0,1),type='b')
```

```
plot(cumsum(pve),xlab="Principal Component", ylab="Cumulative Proportion of variance  
Explained" ,ylim=c(0,1), type='b')
```

```
tg<-predict(pr.out,trainingSet)
```

```
ts<-predict(pr.out,testingSet)
```

```
tg<-data.frame(trainingSet[6],tg)
```

```
ts<-data.frame(testingSet[6],ts)
```

```
#####LDA#####
```

```
lda.fit<-lda(Dummy~PC1+PC2, data=tg)
```

```
lda.fit
```

```
plot(lda.fit)
```

```
lda.pred<-predict(lda.fit, ts)
```

```
names(lda.pred)
```

```
lda.class=lda.pred$class
```

```
length(ts$Dummy)
```

```
length(lda.class)
```

```
table(lda.class, ts$Dummy)
```

```
mean(lda.class==ts$Dummy)
```

```
sum(lda.pred$posterior[,1]>=.5)
```

```
sum(lda.pred$posterior[,1]<.5)
```

```
lda.pred$posterior[1:10,1]
```

```
lda.class[1:10]
```

```
sum(lda.pred$posterior[,1]>.9)
```

```
#####Logistic Regression#####Logistic Regression
```

```
setwd(choose.dir())
```

```
library(ISLR)
```

```
library(dplyr)
```

```
Iowa_IVR_Cases<- read.csv("selectedIowaVariables.csv" ,header=TRUE, sep=",")
```

```
dim(Iowa_IVR_Cases)
```

```
IVR <- Iowa_IVR_Cases %>% mutate_if(is.character, as.factor)
```

```
str(IVR)
```

```
summary(IVR)
```

```
dim(IVR)
```

```
names(IVR)
```

```
contrasts(IVR$Employed_Closure)
```

```
#NO 0
```

```
#YES 1
```

```
#####
```

```
glm.fits<-glm(IVR$Employed_Closure~Disability_Type, data=IVR, family=binomial)
```

```
summary(glm.fits)
```

```
# pvalues of Gender indicate there
```

```
#is no clear evidence of a real
```

```
#association between Gender and
```

```
#Employed_Closure
```

```
glm.probs<-predict(glm.fits, type="response")
```

```
glm.probs[1:10]
```

```
glm.pred<-rep("NO",54593)
```

```
glm.pred[glm.probs>.5]="YES"
```

```
table(glm.pred,IVR$Employed_Closure)
```

```
(10305+21110)/54593
```

```
mean(glm.pred==IVR$Employed_Closure)
```

```
#####
```

```
s_train<-(IVR$Dummy_Emp>0)
```

```

##test#####

NotEmp=IVR[!s_train,]

dim(NotEmp)

summary(NotEmp)

glm3=glm(Dummy_Emp~Disability_Type, data=IVR,
family=binomial, subset=s_train)
glm.probs=predict(glm.fits,NotEmp,type="response")

#####

glm.pred2=rep("0",25208)
glm.pred2[glm.probs>.5] = "1"
table(glm.pred2,NotEmp$Dummy_Emp)
mean(glm.pred2==NotEmp$Dummy_Emp)
mean(glm.pred2!=NotEmp$Dummy_Emp)

summary(IVR$Disability_Type)

##Cognitive 20932
##Deaf & Hard of Hearing 3059
##Psychosocial Mental 18557
##Visual & Communicative 427
## Physical 11618

11618 +427+18557+3059+20932

```

```
#####Train
set.seed(777)
Sampled<-sample(2,nrow(IVR), replace=TRUE, prob=c(0.8,0.2))
trainingSet<-IVR[Sampled==1,]

dim(trainingSet)
dim(IVR)

#####create tests

test1 <- trainingSet %>%
filter(Disability_Type=="Physical")
dim(test1)

test2 <- trainingSet %>%
filter(Disability_Type=="Cognitive")
dim(test2)

test3 <- trainingSet %>%
filter(Disability_Type=="Deaf & Hard of Hearing")
dim(test3)

test4 <- trainingSet %>%
filter(Disability_Type=="Psychosocial Mental")
dim(test4)
```

```

test5 <- trainingSet %>%
  filter(Disability_Type=="Visual & Communicative")
dim(test5)

dim(IVR)
11618/54593

#####testing#####

logitEmp=glm(Dummy_Emp~Disability_Type,data=IVR,family=binomial)
glm.exp1<-predict(logitEmp, newdata=test1,type="response")
glm.exp1[1:10]

glm.p1<-rep("0",9340)
glm.p1[glm.exp1>.5]="1"

table(glm.p1,test1$Dummy_Emp)
mean(glm.p1==test1$Dummy_Emp)
mean(glm.p1!=test1$Dummy_Emp)

####test2

#logitEmp=glm(Dummy_Emp~Disability_Type,data=IVR,family=binomial)
glm.exp2<-predict(logitEmp, newdata=test2,type="response")

```

```
glm.exp2[1:10]
```

```
glm.p2<-rep("0",16803)
```

```
glm.p2[glm.exp2>.5]="1"
```

```
table(glm.p2,test2$Dummy_Emp)
```

```
mean(glm.p2==test2$Dummy_Emp)
```

```
mean(glm.p2!=test2$Dummy_Emp)
```

```
##test3
```

```
#logitEmp=glm(Dummy_Emp~Disability_Type,data=IVR,family=binomial)
```

```
glm.exp3<-predict(logitEmp, newdata=test3,type="response")
```

```
glm.exp3[1:10]
```

```
glm.p3<-rep("0",2485)
```

```
glm.p3[glm.exp3>.5]="1"
```

```
table(glm.p3,test3$Dummy_Emp)
```

```
mean(glm.p3==test3$Dummy_Emp)
```

```
mean(glm.p3!=test3$Dummy_Emp)
```

```
#test4
```

```
#logitEmp=glm(Dummy_Emp~Disability_Type,data=IVR,family=binomial)
```

```
glm.exp4<-predict(logitEmp, newdata=test4,type="response")
```

```
glm.exp4[1:10]
```

```
glm.p4<-rep("0",14818)
```

```
glm.p4[glm.exp4>.5]="1"
```

```
table(glm.p4,test4$Dummy_Emp)
```

```
mean(glm.p4==test4$Dummy_Emp)
```

```
mean(glm.p4!=test4$Dummy_Emp)
```

```
##test 5
```

```
#logitEmp=glm(Dummy_Emp~Disability_Type,data=IVR,family=binomial)
```

```
glm.exp5<-predict(logitEmp, newdata=test5,type="response")
```

```
glm.exp5[1:10]
```

```
glm.p5<-rep("0",338)
```

```
glm.p5[glm.exp5>.5]="1"
```

```
table(glm.p5,test5$Dummy_Emp)
```

```
mean(glm.p5==test5$Dummy_Emp)
```

```
mean(glm.p5!=test5$Dummy_Emp)
```

```
#Job after case closed
```

```
mean(glm.p1==test1$Dummy_Emp) #Physical
```

```
# 0.5185225
```

```

mean(glm.p2==test2$Dummy_Emp) #Cognitive
# 0.5976314
mean(glm.p3==test3$Dummy_Emp) #Deaf & Hard of Hearing
#0.7521127
mean(glm.p4==test4$Dummy_Emp) #Psychosocial Mental
# 0.5508841
mean(glm.p5==test5$Dummy_Emp) #Visual & Communication
#0.612426

#NO JOB after case closed:
mean(glm.p1!=test1$Dummy_Emp) #Physical
#0.4814775
mean(glm.p2!=test2$Dummy_Emp) #Cognitive
# 0.4023686
mean(glm.p3!=test3$Dummy_Emp) #Deaf & Hard of Hearing
#0.2478873
mean(glm.p4!=test4$Dummy_Emp) #Psychosocial Mental
#0.4491159
mean(glm.p5!=test5$Dummy_Emp) #Visual & Communication
#0.387574

#####Forecasting#####

#####select working directory

#setwd(choose.dir())
EmployedPop<-read.csv("employedPop.csv",header=TRUE, sep=",")
summary(EmployedPop)
names(EmployedPop)
attach(EmployedPop)

```

```
plot(Year,Employed_Disability_Pop, type='l',xlab='Year', ylab='Employed Population')
```

```
plot(Year,Employed_Nondisabled_Pop, type='l',xlab='Year', ylab='Non-disabled Workers In  
United States(Number in Thousands)')
```

```
#####
```

```
plot(Year,D_Average, type='l',xlab='Year', ylab='Disabled Workers In United States(Number in  
Thousands)')
```

```
plot(Year,Average, type='l',xlab='Year', ylab='Non-disabled Workers In United States(Number  
in Thousands)')
```

```
#####
```

```
DisaEmpPop<-EmployedPop[,-1][,-3]
```

```
names(DisaEmpPop)
```

```
library(forecast)
```

```
y=EmployedPop[,5]
```

```
names(y)
```

```
y
```

```
m<-EmployedPop[,4]
```

```
x1<-
```

```
c(2009,2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020,2021,2022,2023,2024,2025  
,2026,2027,2028,2029,2030)
```

```
e<-ses(m,h=8)
```

```
de<-ses(y, h=8)
```

```
#disabled
```

```
round(accuracy(de),2)
```

```
summary(de)
#not disabled
round(accuracy(e),2)
summary(e)
#####
library(ggplot2)
summary(de)
names(de)

autoplot(de) +
  autolayer(fitted(de), series="Fitted") +
  ylab("Disabled Workers In United States(Number in Thousands)") + xlab("Year")

autoplot(e) +
  autolayer(fitted(e), series="Fitted") +
  ylab("Non-disabled Workers In United States(Number in Thousands)") + xlab("Year")
```